

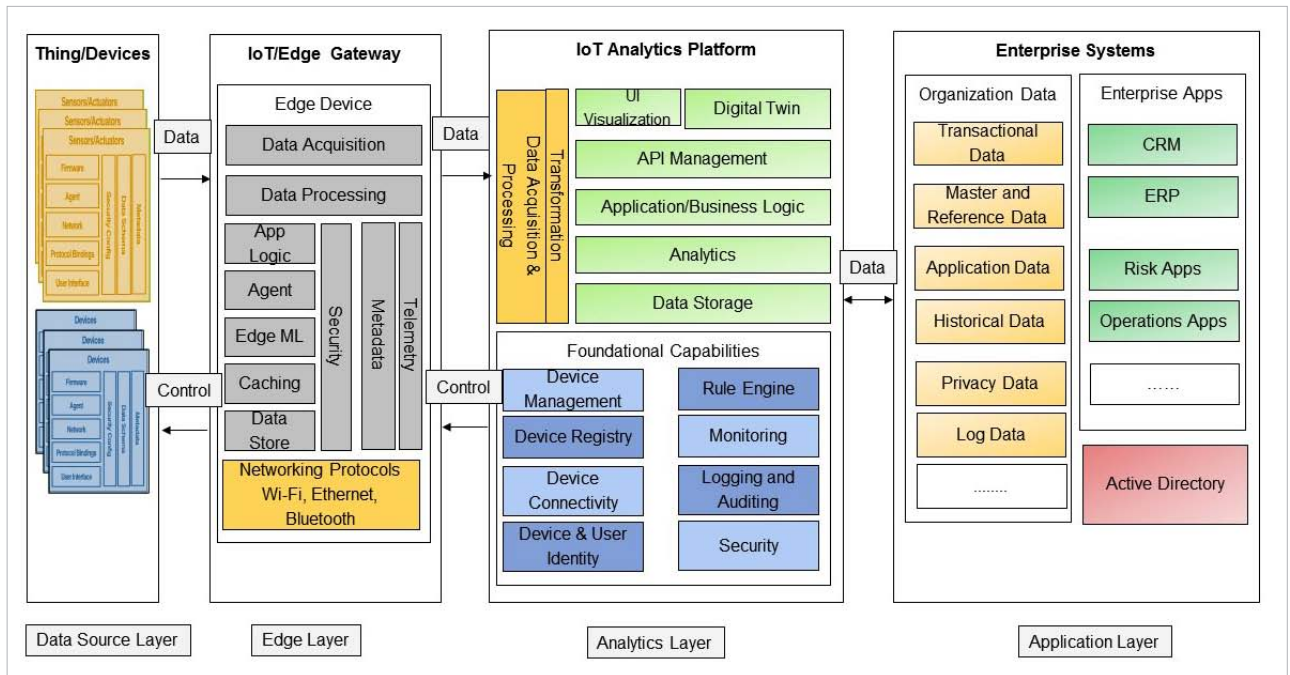


Figure 1: IoT analytics logical reference architecture

It consists of an IoT gateway that facilitates bi-directional communication with the platform and enterprise systems. It also implements edge intelligence with different processing capabilities, enabling connectivity between devices, routing, filtering, and protocol and identity translation.

Analytics layer: The primary purpose of this layer is to interface with the edge layer to receive, store, process, and analyse the data. In addition to data acquisition capabilities, it offers AI-driven analytics for intelligence and visualisation, device management, security, connectivity and network management, and operations.

The data from devices is collected and aggregated from the IoT device gateway. Real-time AI-driven analytics and visualisation combined with machine and deep learning models are applied to the data to derive intelligence.

Machine learning is also performed over the collected device telemetry data, and this data is blended with other enterprise data or secondary sources of telemetry from other systems or organisations.

Application layer: This layer acts as an interface to interact with applications for end users. Devices like smartphones, computers and tablets are used as a medium to make IoT applications interactive.

Real world use cases of IoT analytics

Implementation of IoT analytics based solutions helps enterprises eliminate redundant assets and products, leading to better operational efficiency and cost savings.

IoT for digital government:

The vision of any government in the digital era is to develop connected and intelligent IoT based systems that contribute to the economy, improve citizen satisfaction, and lead to a safer society and environment sustainability.

Processing of data feeds from various sources like weather and environmental sensors, traffic signal cameras, parking zones, etc, leads to improved government-IoT agency coordination and better services to citizens.

IoT for manufacturing: IoT in manufacturing allows richer insights in

real-time. It can facilitate the production flow in a manufacturing plant, as IoT devices automatically monitor development cycles and manage inventories.

IoT for supply chain

management: Manufacturers use IoT to monitor their supply chain and get estimates of the available resources. This includes information regarding the status of the work on the floor, equipment collection, and the delivery date of required materials.

IoT for smart homes: IoT helps to analyse and optimise use of devices in a smart home. This can mean simple switching off or dimming of lights, and modifying multiple home settings to optimise energy use. It also helps discover problematic energy consumption by older and damaged appliances, or faulty system components.

IoT for media: IoT based applications in media and advertising lead to a customised experience in which the system analyses and responds to the needs and interests of each customer. It helps in understanding the customer behaviour